5 A curve is defined implicitly by the equation $2x^2 + 3xy + y^2 + 2 = 0$.

(a) Show that $\frac{dy}{dx} = -\frac{4x + 3y}{3x + 2y}$. **[3]**

(b) In this question you must show detailed reasoning.

Find the coordinates of the stationary points of the curve. **[4]**